Identification of an Unknown – Sample B

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Introduction

The identity of a sample of the unknown structure was deduced by analyzing Nuclear Magnetic Resonance (NMR), Infrared Spectroscopy (FTIR), and Gas Chromatography-Mass Spectrometry (GC-MS) data obtained. The ability to identify unknown compounds can be used to ensure the product of a reaction is the intended end product.

Theory

Nuclear Magnetic Resonance (NMR) is a method of chemical analysis that utilizes electromagnetic radiation in the form of radio waves concerning the protons (hydrogen atoms) in a compound and the impact of the electrons on them. The nuclei of the hydrogen atoms act as a small magnetic force that interacts with an external magnetic field generated by the NMR instrument. When this field is generated, it excites the protons, pushing them into a higher energy state; as the protons return to their normal energy state, the decreasing frequency of waves is used to interpret critical characteristics of the compound. Wave interference, such as destructive and constructive interference, can impact the signal received by canceling out or increasing the intensity of the signal observed. When analyzing ^{1}H NMR Spectra, four main components should be noted, including the number of groups of signals, the integration number of each group, the chemical shift, and patterns within groups of signals. The number of groups of signals is the number of *different kinds* of protons in the compound, and the integration number of each group represents the number of that specific type of proton in the compound. The location of the signal on an NMR spectrum is the chemical shift, which is reported in parts per million (ppm) and utilizes the reference compound, tetramethylsilane (TMS), to compare the absorption frequency of the sample compound. Finally, pattern recognition can help quickly identify substituents and structural features by recognizing patterns indicative of specific attributes.

However, it should be noted that certain patterns can be hidden or artificially made by interference of signals, so this should not be the sole basis on which the chemical structure is determined.

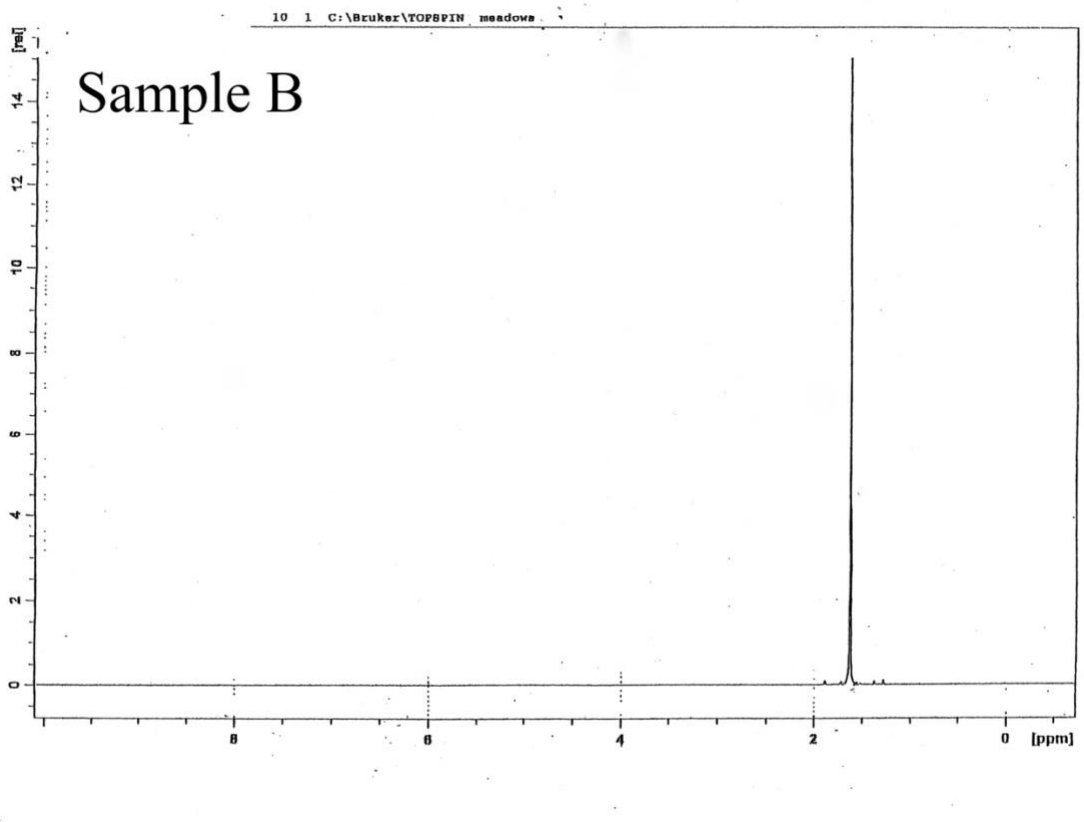
Fourier Transform Infrared (FT-IR) Spectroscopy is a method of chemical analysis that passes a beam of heat radiation energy through a sample and measures the radiation that is not absorbed by the sample to determine the actual absorption. The absorption spectrum of FTIR is plotted with the wavenumber on the x-axis against the percent transmittance on the y-axis. When deciphering the absorption spectrum of FTIR, two central regions can be divided into the diagnostic and fingerprint regions; the diagnostic region has a wavenumber greater than 1500 cm^{-1} and contains signals that reveal double bonds, triple bonds, and halogen-proton bonds. The fingerprint region has a wavenumber less than 1500 cm^{-1} that contains signals resulting from single bonds. The intensity of a signal on the IR spectrum can indicate the sample's polarity in a particular region. The higher the intensity of the peak, the more polar it is assumed to be, and the weaker the intensity of the peak, the more non-polar it is considered to be. This is because the dipole moment created by a compound oscillates as a function of time, and the strength of this dipole moment will extend the time of oscillation and, subsequently, the intensity of the signal obtained. The shape of the signals on the absorption spectrum is affected by various factors, such as the type of bonding and the symmetry (or lack thereof). Signals of functional groups involved in hydrogen bonding are broader and span farther than those considered "free" of hydrogen bonding. This is due to the weakening effect on each group, resulting in an uneven bond strength distribution. The symmetry of a functional group can impact the number of signals observed due to the different ways the group can stretch or vibrate. In symmetrical compounds, there is only one way that the signal can be received, no matter which way it vibrates due to its symmetry. In

contrast, asymmetric compounds can result in two or more signals that vary based on the vibrational differences. The attributes outlined above make FTIR spectroscopy a valuable tool for identifying potential functional groups in a sample.

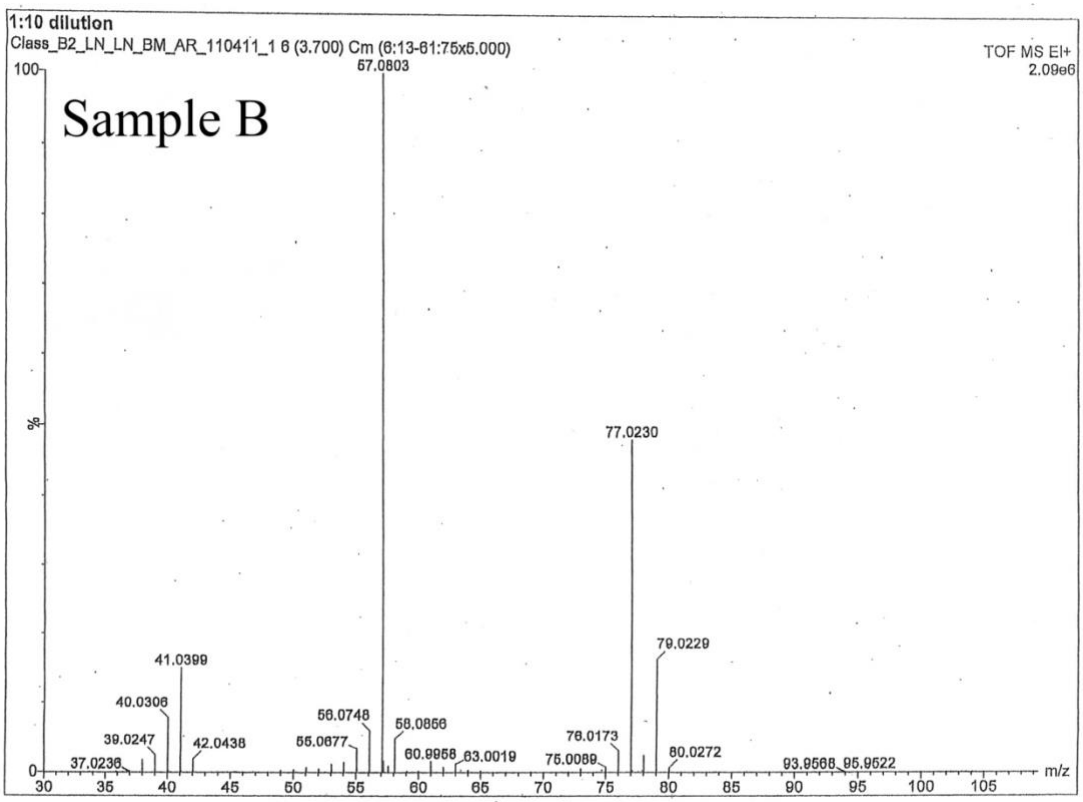
Gas Chromatography & Mass Spectrometry (GC-MS) are used in tandem to optimize the efficiency and accuracy of identifying unknown compounds. Gas chromatography separates the individual components of a compound as it travels through the column based on their chemical properties. The separated components are introduced to the mass spectrometer as gas-phase ions and then subjected to electron impact via high-energy electrons. This generates a radical cation intermediate, termed the molecular ion, and results in fragmentation due to radical ion instability. The cationic fragments and molecular ions are deflected by the magnetic field and separated by their mass-to-charge ratio (m/z). The mass spectrum generated depicts each cation's relative abundance (%). When analyzing an electron impact spectrum, the molecular ion peak is typically assumed to be the peak with the highest mass-to-charge ratio since the molecular ion is the least fragmented version of the component. The most abundant component in the structure is considered the base peak and will have a percent abundance of 100%, serving as a relative point for the percent abundance of the other fragments. Additional considerations for EI spectrum analysis include relative isotope abundances and the presence of halogens such as chlorine and bromine. Isotopic abundances can impact the intensity of signals, where the M signal can also be accompanied by the isotopic signal at +1 or +2 m/z, dependent on relative abundance. This can be seen when the molecular ion is chlorine or bromine; chlorine will give a characteristic molecular ion peak that has an M+2 peak that is approximately one-third the size of the molecular ion peak. Bromine will also have a molecular ion peak accompanied by an M+2 peak, but the M+2 peak will be almost equivalent to the molecular ion peak.

Data

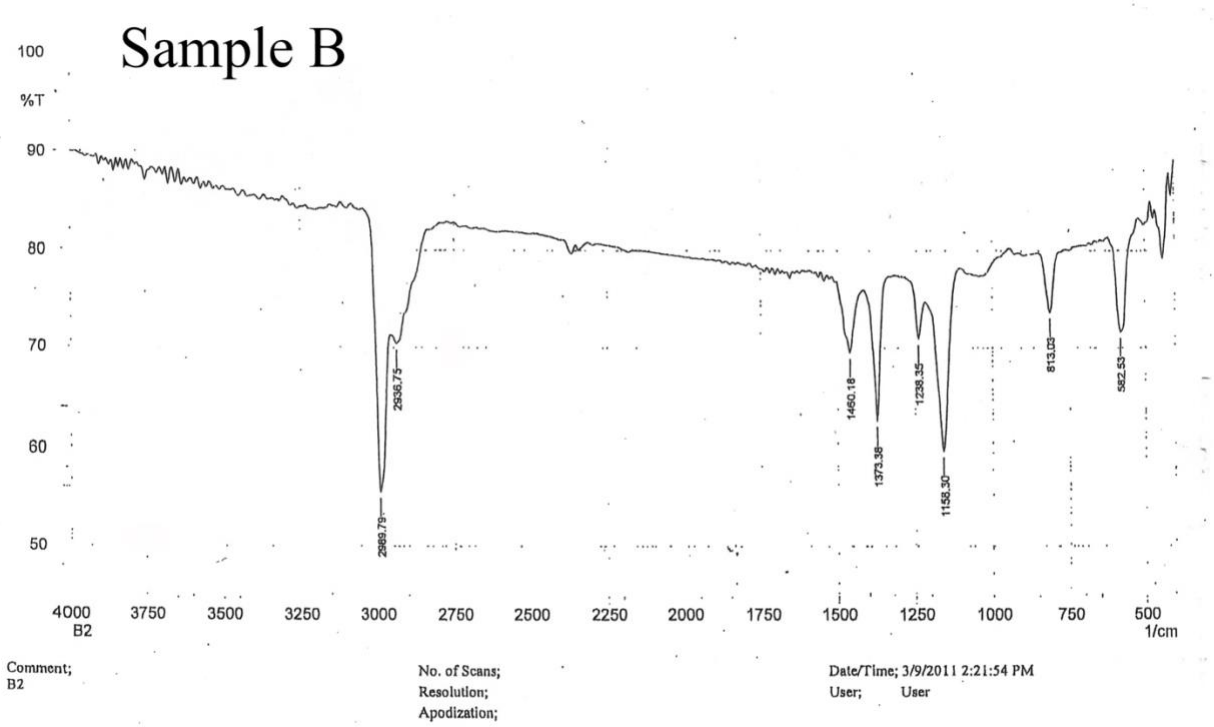
^{1}H NMR


10 1 C:\Bruker\TOPSPIN meadowa
Sample B
[rel]
14
12
10
8
6
4
2
0
8
6
4
2
0
[ppm]

GC-MS


1:10 dilution
Class_B2_LN_LN_BM_AR_110411_1 6 (3.700) Cm (6:13-61:75x5.000)
TOF MS EI+
2.09e6
Sample B
100
%
0
57.0803
77.0230
79.0229
41.0399
40.0306
56.0748
58.0656
39.0247
42.0438
55.0677
76.0173
80.0272
60.9958
63.0019
75.0089
37.0236
93.9568
95.9522
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105

FTIR



Analysis

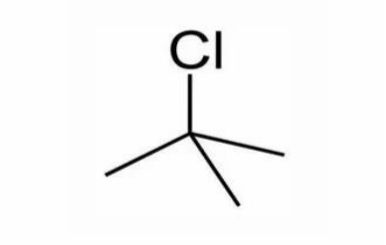
The NMR spectrum contained one signal, a high-intensity singlet at approximately 1.7 ppm. The lack of additional signals indicates that the sample has only one kind of proton, meaning that all protons in this structure must be chemically and magnetically equivalent to one another. The isolated singlet in the 1.4 – 1.9 ppm range is a characteristic pattern of the presence of a tert-butyl group, particularly with an integration number of 9. When assessing the GC-MS spectrum, it was noted that the base peak has a mass-to-charge ratio of approximately 57.1,

indicating the presence of a C_4H_9 radical. This, along with the characteristic tert-butyl signal on the NMR spectrum, aids the assumption that a tert-butyl group is part of the unknown compound. The single signal of the NMR also reveals that the unknown compound is symmetrical due to the chemical equivalence of the protons associated with the compound. Upon further analysis of the GC-MS spectrum, the molecular ion was identified as having a mass-to-charge ratio of roughly 77.0. However, a noticeable peak with a mass-to-charge ratio of 79.0 indicates a possible isotope. This pattern is characteristic of an M+2 peak pattern; the M+2 peak had an abundance that was approximately one-third of the molecular ion's peak; this is typically seen in compounds that contain chlorine, as chlorine has two main isotopes found in organic compounds (^{35}Cl & ^{37}Cl), with relative abundances of 100 % to 32%, respectively. The observed M+2 peak pattern exhibits the isotopes' relative abundance to one another at a 3:1 ratio. Therefore, it can be deduced that a chlorine atom is present in the unknown structure. Analysis of the FTIR absorption spectrum revealed no significant signals in 1850 cm^{-1} – 1630 cm^{-1}, 3550 cm^{-1} – 3200 cm^{-1}, or 2280 cm^{-1} – 2100 cm^{-1} regions; this indicates the absence of carbonyl groups, hydroxyl groups, primary & secondary amines, and cyanide groups respectively. Strong peaks in the 2990 cm-1 – 2850 cm-1 region indicate vibrational stretching of C-H bonds in alkyl groups, and multiple strong/medium peaks in the 1500 cm^{-1} – 1000 cm^{-1} region indicate numerous methyl groups. Finally, the peak at the 582 cm^{-1} point supports the GC-MS spectrum's data, indicating the presence of a chlorine atom.

Conclusion

The ^1H NMR, FTIR, and GC-MS spectrum data indicate the presence of a tert-butyl group, a chlorine atom, and a symmetrical structure consisting of chemically and magnetically equivalent protons. After ruling out additional structures through FTIR analysis, it can be

deduced that the unknown sample is 2-chloro-2-methylpropane (*tert-Butyl chloride*). The chlorine atom would have to be attached directly to the central carbon atom of the tert-butyl group to maintain the symmetry of the structure and chemical equivalence of the protons.



2-chloro-2-methylpropane

References

Keller L, Joseph K, and Miguel S. 2023 Fall. *CHEM 2081 Organic Chemistry II Lab Manual* [lab manual]. Auburn (Al): Auburn University.

Klein, D. R. (2022). *Organic Chemistry* (Fourth). John Wiley & Sons. November 25, 2023, https://platform.virdocs.com/read/1892025/1/#/4/2[c08-sec-0001]/2